Effects of sotatercept on haemodynamics and right heart function: analysis of the STELLAR trial

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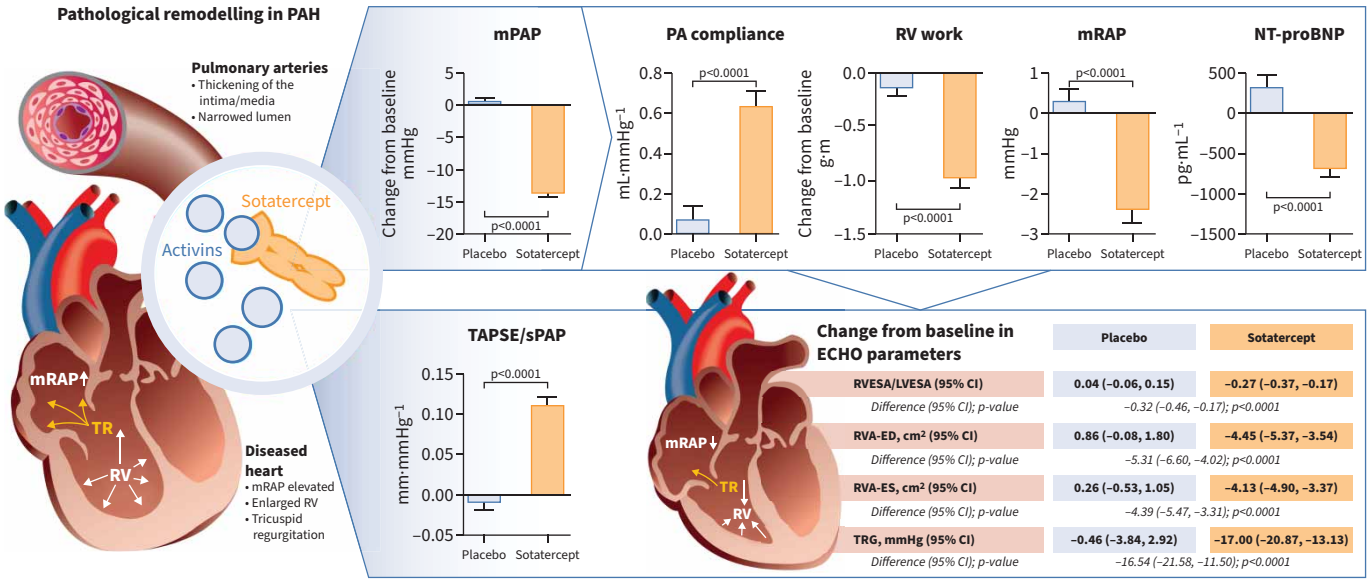

GRAPHICAL ABSTRACT Summary of the main study findings. Treatment with sotatercept for 24 weeks significantly reduced right heart size and improved right ventricle (RV) function and haemodynamic status in patients with pulmonary arterial hypertension (PAH). Error bars in graphs represent standard error. ECHO: echocardiography; LVESA: left ventricle end-systolic area; mPAP: mean pulmonary artery pressure; mRAP: mean right atrial pressure; NT-proBNP: N-terminal pro-brain natriuretic peptide; PA: pulmonary artery; RVA-ED: right ventricular area in end-diastole; RVA-ES: right ventricular area in end-systole; RVESA: right ventricle end-systolic area; sPAP: systolic pulmonary artery pressure; TAPSE: tricuspid annular plane systolic excursion; TRG: tricuspid regurgitation gradient.

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Abstract

Background In the phase 3 STELLAR trial, sotatercept, an investigational first-in-class activin signalling inhibitor, demonstrated beneficial effects on 6-min walk distance and additional efficacy endpoints in pre-treated participants with pulmonary arterial hypertension (PAH).

Methods This *post hoc* analysis evaluated data from right heart catheterisation (RHC) and echocardiography (ECHO) obtained from the STELLAR trial. Changes from baseline in RHC and ECHO parameters were assessed at 24 weeks. An analysis of covariance (ANCOVA) model was used to estimate differences in least squares means with treatment and randomisation stratification (mono/double *versus* triple therapy; World Health Organization functional class II *versus* III) as fixed factors, and baseline value as covariate.

Results Relative to placebo, treatment with sotatercept led to significant (all $p<0.0001$ except where noted) improvements from baseline in mean pulmonary artery (PA) pressure (−13.9 mmHg), pulmonary vascular resistance (−254.8 $dyn \cdot s \cdot cm^{-5}$), mean right atrial pressure (−2.7 mmHg), mixed venous oxygen saturation (3.84%), PA elastance (−0.42 $mmHg \cdot mL^{-1} \cdot beat^{-1}$), PA compliance (0.58 $mL \cdot mmHg^{-1}$), cardiac efficiency (0.48 $mL \cdot beat^{-1} \cdot mmHg^{-1}$), right ventricular (RV) work (−0.85 g·m) and RV power (−32.70 $mmHg \cdot L \cdot min^{-1}$). ECHO showed improvements in tricuspid annular plane systolic excursion (TAPSE) to systolic pulmonary artery pressure ratio (0.12 $mm \cdot mmHg^{-1}$), end-systolic and end-diastolic RV areas (−4.39 cm^2 and −5.31 cm^2, respectively), tricuspid regurgitation and RV fractional area change (2.04% $p<0.050$). No significant between-group changes from baseline were seen for TAPSE, heart rate, cardiac output, stroke volume or their indices.

Conclusion In pre-treated patients with PAH, sotatercept demonstrated substantial improvements in PA pressures, PA compliance, PA–RV coupling and right heart function.

Introduction

Pulmonary arterial hypertension (PAH) is a disease characterised by progressive luminal narrowing of the small pulmonary arteries with abnormal elevations in pulmonary artery pressure. The chronic pressure overload initially leads to adaptive and later maladaptive remodelling of the right heart, involving pathological tissue changes characterised by hypertrophy and dilatation of both the right atrium (RA) and right ventricle (RV), tricuspid regurgitation (TR), altered RV–pulmonary artery (PA) coupling, and impaired RV contractility [1]. The symptoms experienced by patients with PAH are mainly related to right heart dysfunction, and right-sided heart failure is the most common cause of death in this patient population [2].

Even with multiple therapeutic advances over recent decades, PAH remains incurable and continues to be associated with high morbidity and mortality, highlighting the need for novel treatments that target the underlying disease process. Research has highlighted the role of altered signal transduction by members of the transforming growth factor β superfamily, including bone morphogenic protein receptor type II, activin receptor type II A (ActRIIA), and the ActRIIA ligands activin A, activin B, growth differentiation factor 8 (GDF8) and GDF11 [3–5]. An imbalance in anti-proliferation signals *via* ActRIIA skews the cellular environment in the vessel wall toward hyperproliferation. Moreover, activin itself plays key roles in several other biological processes beyond fibrosis, such as inflammation and angiogenesis, which also are implicated in PAH [6].

Sotatercept, a first-in-class, investigational activin signalling inhibitor, is a soluble fusion protein comprised of the extracellular domain of ActRIIA linked to the Fc portion of human IgG1. By selectively binding and trapping ligands of the ActRIIA and similar receptors, sotatercept aims to restore the balance between pro-proliferative and anti-proliferative signalling in PAH [7]. While direct clinical evidence is yet to be confirmed, animal studies suggest that sotatercept inhibits cellular proliferation, promotes apoptosis, and alleviates inflammation in pulmonary vessel walls, leading to reverse remodelling and restoration of vessel patency [5, 8, 9].

In the phase 2 PULSAR trial and the phase 3 STELLAR trial, both conducted in pre-treated patients with group 1 PAH and symptomatic World Health Organization (WHO) functional class II or III disease, sotatercept treatment was superior to placebo in improving pulmonary vascular resistance (PVR) and exercise capacity (assessed by 6-min walk distance (6MWD)). A substudy from PULSAR showed improvements in echocardiographic indices of right heart function [10]. In STELLAR, sotatercept delivered broad clinical benefit across multiple efficacy endpoints, including haemodynamics, WHO functional class, N-terminal pro-brain natriuretic peptide (NT-proBNP), risk scores, patient-reported outcomes at week 24, and time to death and non-fatal clinical worsening events [11]. Notably, in both PULSAR and STELLAR, sotatercept had unique haemodynamic effects as PVR was reduced solely by a drop in mean PA pressure (mPAP), while cardiac output remained unchanged. This differs from currently approved PAH medications, which usually cause a decrease in PVR by reducing mPAP and increasing cardiac output [10–16]. To better understand the effects of sotatercept on haemodynamics and cardiac function, we performed an in-depth *post hoc* exploratory analysis of right heart catheterisation (RHC) and echocardiography (ECHO) findings from the STELLAR trial.

Materials and methods

Study subjects

The details of the STELLAR trial are described elsewhere [11]. Briefly, the STELLAR trial (ClinicalTrials.gov NCT04576988) enrolled 323 participants with group 1 PAH and symptomatic WHO functional class II or III disease, PVR ⩾400 dyn·s·cm^{-5} and 6MWD 150–500 m who were on stable background PAH therapies. Participants had a confirmed diagnosis of PAH (idiopathic, heritable, drug-induced, connective-tissue disease-associated or following shunt correction), excluding subtypes associated with portopulmonary disease, schistosomiasis, HIV infection, or veno-occlusive disease. All participants received stable background therapy for PAH for ⩾90 days prior to enrolment and continued the same therapy throughout the trial. Background treatments consisted of monotherapy, double therapy or triple therapy with currently available medications for PAH.

Study design and measurements

Eligible participants were stratified by baseline WHO functional class (II *versus* III) and background therapy (monotherapy/double therapy *versus* triple therapy) and were randomised 1:1 to double-blind treatment with sotatercept (starting dose 0.3 mg·kg^{-1} escalated to target dose 0.7 mg·kg^{-1}) or placebo administered subcutaneously every 3 weeks in combination with background therapies. RHC and ECHO were performed at baseline and week 24. Per protocol, concomitant PAH therapies were to be kept stable during this time period. RHC and ECHO variables assessed in the present analysis are listed in

TABLE 1 Definition of right heart catheterisation parameters, unit of measurement and measurement method

Abbreviation	Parameter	Description	Units	Measurement method
SBP	Systolic blood pressure	Peak systemic blood pressure	mmHg	Directly measured
DBP	Diastolic blood pressure	Lowest systemic blood pressure	mmHg	Directly measured
SVR	Systemic vascular resistance	Resistance to blood flow in the systemic circulation	$dyn \cdot s \cdot cm^{-5}$	Calculated: 80×((2×DBP + SBP)/3 – RAP)/CO
PVR	Pulmonary vascular resistance	Resistance to blood flow in the pulmonary vasculature	$dyn \cdot s \cdot cm^{-5}$	Calculated: (mPAP–PAWP)/CO
mPAP	Mean pulmonary artery pressure	Average pressure within the pulmonary artery	mmHg	Directly measured
sPAP	Systolic pulmonary artery pressure	Peak pressure in the pulmonary artery during systole	mmHg	Directly measured
dPAP	Diastolic pulmonary artery pressure	Lowest pressure in the pulmonary artery during diastole	mmHg	Directly measured
PAWP	Pulmonary artery wedge pressure	Pressure in the pulmonary artery after proximal balloon occlusion	mmHg	Directly measured
mRAP	Mean right atrial pressure	Pressure within the right atrium	mmHg	Directly measured
CO	Cardiac output	Volume of blood pumped by the heart per minute	$L \cdot min^{-1}$	Measured in this study by thermodilution or the indirect Fick technique
CI	Cardiac index	Cardiac output normalised to body surface area	$L \cdot min^{-1} \cdot m^{-2}$	Calculated: CO/body surface area
HR	Heart rate	Number of heart beats per minute	$beats \cdot min^{-1}$	Directly measured
SV	Stroke volume	Volume of blood pumped by the heart per beat	$mL \cdot beat^{-1}$	Calculated: CO/HR
SVI	Stroke volume index	Stroke volume normalised to body surface area	$mL \cdot beat^{-1} \cdot m^{-2}$	Calculated: SV/body surface area
CE	Cardiac efficiency	Ratio of stroke volume to mPAP	$mL \cdot mmHg^{-1}$	Calculated: SV/mPAP
PA compliance	Pulmonary artery compliance	Measure of the ability of the pulmonary artery to expand and contract	$mL \cdot mmHg^{-1}$	Calculated: SV/(sPAP–dPAP)
RC product	Resistance–compliance product	The product of resistance and compliance of the pulmonary arteries	s	Calculated: PA compliance × PVR
PA elastance	Pulmonary artery elastance	Measure of right ventricular afterload	$mmHg \cdot mL^{-1}$	Calculated: sPAP/SV
RV work	Right ventricular work	Measure of right ventricular workload	g·m	Calculated: CO × mPAP × 0.0144
RV power	Right ventricular power	Measure of right ventricular workload	$mmHg \cdot L \cdot min^{-1}$	Calculated: mPAP × CI
S_{vO_2}	Mixed venous oxygen saturation	Oxygen saturation in blood obtained from the pulmonary artery	%	Directly measured

tables 1 and 2, respectively. RHC data were obtained at the centres while ECHO was locally recorded but centrally exploited in a core lab (Biotelemetry, Malvern, PA, USA). NT-proBNP was assessed at baseline and every 3 weeks thereafter, including week 24.

Analysis

Baseline demographics and disease characteristics were summarised using descriptive statistics for the cohort of participants with paired baseline and week 24 measurements. An analysis of covariance (ANCOVA) model was used to estimate differences in least squares (LS) means with treatment (sotatercept *versus* placebo) and randomisation stratification factors (mono/double therapy *versus* triple therapy and WHO functional class II *versus* III) as fixed factors, and baseline value for the parameter under evaluation as a covariate. For continuous parameters, within-group LS mean changes and between-group differences in LS means at week 24 were calculated for each parameter along with standard deviations, 95% confidence intervals and p-values. For the categorical variable tricuspid regurgitation degree, Fisher's exact test was used to compare the distribution of categories between the treatment groups at week 24. All tests were two-sided and p-values <0.050 were considered statistically significant. No adjustments were made for multiple testing.

For correlation analyses, Spearman's rank-order coefficients were computed to evaluate the relationships between changes from baseline at week 24 in the RHC and ECHO parameters as well as NT-proBNP or 6MWD for the overall population. For these analyses, NT-proBNP was log_2 transformed to normalise the distribution of values, stabilise the variance and render the data suitable for analysis. The null hypothesis

TABLE 2 Definition of echocardiography parameters, unit of measurement and measurement method

Acronym	Parameter	Description	Units	Measurement method
sPAP	Systolic pulmonary artery pressure	Peak pressure in the pulmonary artery during systole	mmHg	Calculated: TRG+eRAP
eRAP	Estimated right atrial pressure	Pressure within the right atrium	mmHg	Estimated based on inferior vena cava diameter and collapsibility
RVFAC	Right ventricular fractional area change	Measure of right ventricular systolic function	%	Calculated: (RVEDA−RVESA)/RVEDA
RVA-ED	Right ventricular area in end-diastole	Area of the right ventricular cavity at end-diastole	cm^2	Directly measured
RVA-ES	Right ventricular area in end-systole	Area of the right ventricular cavity at end-systole	cm^2	Directly measured
RVEDA/LVEDA	Ratio of right ventricular end-diastolic to left ventricular end-diastolic area	Measure of right ventricle enlargement compared to left ventricle during diastole	Unitless	RVEDA/LVEDA
RVESA/LVESA	Ratio of right ventricular end-systolic to left ventricular end-systolic area	Measure of right ventricle enlargement compared to left ventricle during systole	Unitless	RVESA/LVESA
TAPSE	Tricuspid annular plane systolic excursion	Measure of right ventricular systolic function	mm	Directly measured
TAPSE/sPAP	Ratio of TAPSE to sPAP	Measure of RV–PA coupling	$mm \cdot mmHg^{-1}$	TAPSE/sPAP
TRG	Tricuspid regurgitation gradient	Pressure gradient across the tricuspid valve based on the speed of regurgitant jet	mmHg	Directly measured
Peak TRV	Peak tricuspid regurgitation velocity	Maximum velocity of blood regurgitating through the tricuspid valve	$m \cdot s^{-1}$	Directly measured
TR degree	Tricuspid regurgitation degree	Severity of tricuspid valve regurgitation	Categorical: none, trace, mild, moderate, severe	Assessed by central reader based on colour flow mapping
LVEF	Left ventricular ejection fraction	Percentage of blood volume ejected from the left ventricle with each heartbeat	%	Directly measured (Simpson's method)
LVEDVI	Left ventricular end-diastolic volume index	Left ventricular end-diastolic volume (LVEDV) normalised to body surface area	$mL \cdot m^{-2}$	Calculated: LVEDV/body surface area
LVESVI	Left ventricular end-systolic volume index	Left ventricular end-systolic volume (LVESV) normalised to body surface area	$mL \cdot m^{-2}$	Calculated: LVESV/body surface area

assumed that the correlation coefficient (Rho) for each relationship pair would be zero. The sample sizes varied for each correlation analysis, depending on the availability of data for the variables under investigation. Results were given as correlation coefficients (CCs).

The treatment effects on cardiac output, cardiac index, heart rate, stroke volume (SV) and stroke volume index (SVI) were evaluated in subgroups defined by baseline cardiac index values using thresholds of ⩾2.0 and 2.5 $L \cdot min^{-1} \cdot m^{-2}$, respectively. Within-group LS mean changes and between-group differences in LS means at week 24 were calculated using the ANCOVA model for the parameter in question across the different subgroups.

Results

Baseline demographics and disease characteristics

The STELLAR trial enrolled a total of 323 participants with PAH, with 163 randomised to sotatercept and 160 randomised to placebo. The current analysis was conducted in a subset of participants with paired baseline and week 24 RHC (n=298) and ECHO (n=275) measurements for the individual parameters under evaluation. No imputations were made for individuals with missing measurements. The detailed baseline characteristics of the cohorts included in the RHC and ECHO analyses are shown in supplementary tables S1a and S1b, respectively. Reasons for missing values are provided in supplementary tables S8 and S9. The RHC and ECHO cohorts were generally well balanced with respect to baseline demographics and disease characteristics.

Treatment effects on RHC parameters

The changes from baseline to week 24 in RHC parameters are shown in table 3 and figure 1. Relative to placebo, treatment with sotatercept led to significant improvements in haemodynamic status. As previously reported, there were significant between-group reductions from baseline in mPAP (−13.9 mmHg, 95% CI −16.0 to −11.8 mmHg; p<0.001) and PVR (−254.8 dyn·s·cm^{-5}, 95% CI −309.0 to −200.6 dyn·s·cm^{-5}; p<0.0001) [11]. Mean RA pressure declined by −2.7 mmHg (95% CI −3.5 to −1.9 mmHg; p<0.0001) at week 24. The decrease in mPAP resulted in improvements in PA compliance (0.58 mL·mmHg^{-1}, 95% CI 0.39 to 0.76 mL·mmHg^{-1}; p<0.0001), PA elastance (−0.42 mmHg·mL^{-1}·beat^{-1}, 95% CI −0.52 to −0.32 mmHg·mL^{-1}·beat^{-1}; p<0.0001), cardiac efficiency (0.48 mL·beat^{-1}·mmHg^{-1}, 95% CI 0.37 to 0.60 mL·beat^{-1}·mmHg^{-1}; p<0.0001), RV work (−0.85 g·m, 95% CI −1.07 to −0.64 g·m; p<0.0001), and RV power (−32.70 mmHg·L·min^{-1}, 95% CI −40.86 to −24.53 mmHg·L·min^{-1}; p<0.0001). The resistance–compliance product did not change in any of the treatment arms or between groups (table 3).

In the overall analysis population, there were no significant between-group changes in cardiac output, cardiac index, SV and SVI at week 24 (table 3). In participants with a baseline cardiac index <2.5 L·min^{-1}·m^{-2} (n=120), the placebo-corrected change in cardiac index was 0.14 L·min^{-1}·m^{-2} (95% CI −0.03 to 0.31 L·min^{-1}·m^{-2}; p=0.1135). In participants with a baseline cardiac index <2.0 L·min^{-1}·m^{-2} (n=34), the placebo-corrected change in cardiac index was 0.22 L·min^{-1}·m^{-2} (95% CI −0.09 to 0.54 L·min^{-1}·m^{-2}; p=0.1525) (supplementary table S2).

Treatment effects on ECHO parameters

The changes from baseline at week 24 in ECHO parameters are shown in table 4 and figure 2. There was no significant between-group change in tricuspid annular plane systolic excursion (TAPSE). However, significant between-group reductions in systolic PA pressure (sPAP) (−18.52 mmHg, 95% CI −24.07 to −12.96 mmHg; p<0.0001) resulted in an improved TAPSE/sPAP ratio of 0.12 mm·mmHg^{-1} (95% CI 0.09 to 0.16 mm·mmHg^{-1}; p<0.0001). In addition, a significant between-group improvement in RV fractional area change (RVFAC) (2.04%, 95% CI 0.03% to 4.05%; p=0.0462) was seen, favouring sotatercept treatment. RV dimensions were reduced with sotatercept relative to placebo, both in terms of absolute and relative changes compared to left ventricle (LV) dimensions.

The distributions in the frequencies of participants across the various tricuspid regurgitation degree categories were similar in the placebo and sotatercept groups at baseline (p=0.2543 for sotatercept *versus* placebo). At week 24, more participants in the sotatercept *versus* placebo group were rated as having a tricuspid regurgitation degree of "none" (34/141 (24.1%) *versus* 3/133 (2.3%), respectively) or "trace" (74/141 (52.5%) *versus* 45/133 (33.8%), respectively), while fewer participants in the sotatercept *versus* placebo group were rated as having a tricuspid regurgitation degree of "mild" (21/141 (14.9%) *versus* 56/133 (42.1%), respectively), "moderate" (7/141 (5.0%) *versus* 21/133 (15.8%), respectively), or "severe" (1/141 (0.7%) *versus* 4/133 (3.0%), respectively) (p<0.0001 for sotatercept *versus* placebo) (figure 3 and supplementary table S3).

Left heart structure and function remained largely unchanged following sotatercept treatment. There was, however, a small but statistically significant between-group decrease in LV ejection fraction (LVEF) of −2.06% (95% CI −3.79% to −0.33%; p=0.0200) at week 24. Measures of LV volume remained stable with non-significant between-group changes in LV end-diastolic volume index (−0.89 mL·m^{-2}, 95% CI −3.35 to 1.57 mL·m^{-2}; p=0.4777) and LV end-systolic volume index (0.65 mL·m^{-2}, 95% CI −0.47 to 1.76 mL·m^{-2}; p=0.2548).

Correlation analyses: change in NT-proBNP associations with RHC and ECHO parameters

The between-group change from baseline in NT-proBNP at week 24 was −996.74 pg·mL^{-1} (95% CI −1349.06 to −644.42 pg·mL^{-1}; p<0.0001) in the overall analysis cohort based on the ANCOVA model.

Correlations between changes from baseline in RHC and ECHO variables with changes from baseline in $\log_2$ NT-proBNP at week 24 are shown in supplementary tables S4 and S5, respectively. The most notable association was a strong positive correlation between change from baseline in mPAP and change from baseline in $\log_2$ NT-proBNP (CC 0.550) (figure 4). Significant correlations were seen between improvements in $\log_2$ NT-proBNP and improvements in RHC parameters of RHC-derived sPAP (CC 0.477) and diastolic PA pressure (CC 0.511) (supplementary table S2). Improvements in $\log_2$ NT-proBNP were moderately (CC ⩾0.400) correlated with improvements in PVR, PA elastance, RV area in end-systole (figure 4) and cardiac efficiency. Moderate (CC <0.400 but ⩾0.350) correlations also were seen between reductions in $\log_2$ NT-proBNP and improvements in RV area in end-diastole, ECHO-derived sPAP, ratio of RV end-systolic to LV end-systolic area and TAPSE/sPAP ratio.

TABLE 3 Baseline, week 24, and change from baseline values in haemodynamic parameters across the placebo and sotatercept groups

	Placebo (n=160)			Sotatercept (n=163)			Sotatercept *versus* placebo difference in LS means at week 24 (95% CI); p-value
	Baseline Mean±SD (n)	Week 24 Mean±SD (n)	LS mean change from baseline at week 24 (95% CI) (n)	Baseline Mean±SD (n)	Week 24 Mean±SD (n)	LS mean change from baseline at week 24 (95% CI) (n)	
SBP (mmHg)	111.5±14.7 (160)	109.9±15.4 (147)	−1.6 (−3.6, 0.3) (147)	112.4±14.2 (163)	114.7±14.6 (159)	2.5 (0.7, 4.4) (159)	4.1 (1.5, 6.8); p=0.0023
DBP (mmHg)	69.3±9.6 (160)	68.6±10.5 (147)	−0.2 (−1.7, 1.3) (147)	67.5±11.3 (163)	72.4±11.4 (159)	4.6 (3.1, 6.0) (159)	4.8 (2.7, 6.9); p<0.0001
SVR (dyn·s·cm^{-5})	1310.0±355.7 (159)	1326.2±395.2 (144)	36.9 (−21.7, 95.5) (144)	1320.2±425.2 (163)	1454.8±464.9 (154)	154.7 (98.5, 210.9) (154)	117.7 (37.7, 197.8); p=0.0041
PVR (dyn·s·cm^{-5})	745.8±313.5 (160)	779.3±335.1 (141)	28.0 (−11.9, 68.0) (141)	781.4±398.5 (163)	545.8±265.2 (154)	−226.8 (−264.6, −189.0) (154)	−254.8 (−309.0, −200.6); p<0.0001
mPAP (mmHg)	52.2±13.0 (160)	52.8±13.8 (144)	0.3 (−1.2, 1.9) (144)	53.0±14.6 (163)	39.6±12.6 (154)	−13.6 (−15.1, −12.1) (154)	−13.9 (−16.0, −11.8); p<0.0001
sPAP (mmHg)	82.59±19.08 (160)	82.36±21.36 (144)	−0.10 (−2.64, 2.44) (144)	82.96±22.17 (163)	60.92±19.64 (154)	−22.16 (−24.60, −19.73) (154)	−22.06 (−25.53, −18.60); p<0.0001
dPAP (mmHg)	34.66±10.86 (160)	35.28±11.49 (144)	0.27 (−1.11, 1.64) (144)	35.84±13.78 (163)	26.03±10.89 (154)	−9.85 (−11.17, −8.54) (154)	−10.12 (−12.00, −8.24); p<0.0001
PAWP (mmHg)	9.83±3.10 (160)	10.20±3.60 (141)	0.29 (−0.24, 0.82) (141)	9.75±3.20 (163)	9.12±3.50 (154)	−0.71 (−1.21, −0.21) (154)	−1.00 (−1.72, −0.29); p=0.0063
mRAP (mmHg)	8.5±4.6 (159)	8.7±5.0 (144)	0.3 (−0.3, 0.9) (144)	8.0±4.3 (163)	5.7±3.3 (154)	−2.4 (−3.0, −1.9) (154)	−2.7 (−3.5, −1.9); p<0.0001
CO (L·min^{-1})	4.8±1.2 (160)	4.7±1.2 (144)	−0.2 (−0.3, −0.0) (144)	4.9±1.3 (163)	4.8±1.3 (154)	−0.1 (−0.3, 0.0) (154)	0.1 (−0.2, 0.3); p=0.5900
CI (L·min^{-1}·m^{-2})	2.7±0.6 (160)	2.6±0.6 (144)	−0.1 (−0.2, −0.0) (144)	2.7±0.6 (163)	2.6±0.6 (154)	−0.1 (−0.2, 0.0) (154)	0.0 (−0.1, 0.2); p=0.6212
HR (beats·min^{-1})	74.39±12.02 (158)	73.27±12.10 (142)	−1.30 (−2.99, 0.39) (142)	75.54±13.17 (155)	72.81±14.64 (150)	−2.46 (−4.11, −0.80) (150)	−1.15 (−3.48, 1.18); p=0.3302
SV (mL·beat^{-1})	66.7±19.4 (158)	65.9±18.9 (142)	−1.4 (−3.9, 1.1) (142)	65.8±17.4 (155)	67.8±20.6 (150)	0.2 (−2.3, 2.6) (150)	1.6 (−1.9, 5.0); p=0.3760
SVI (mL·beat^{-1}·m^{-2})	36.7±9.9 (158)	36.1±9.9 (142)	−0.8 (−2.2, 0.5) (142)	36.7±8.9 (155)	37.1±9.6 (150)	−0.2 (−1.5, 1.2) (150)	0.7 (−1.1, 2.5); p=0.4596
CE (mL·beat^{-1}·mmHg^{-1})	1.36±0.51 (158)	1.37±0.63 (142)	−0.00 (−0.09, 0.09) (142)	1.34±0.53 (155)	1.88±0.86 (150)	0.48 (0.40, 0.57) (150)	0.48 (0.37, 0.60); p<0.0001
PA compliance (mL·mmHg^{-1})	1.52±0.65 (158)	1.62±0.98 (142)	0.07 (−0.07, 0.20) (142)	1.51±0.59 (154)	2.22±1.02 (150)	0.64 (0.51, 0.78) (150)	0.58 (0.39, 0.76); p<0.0001

Continued

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TABLE 3 Continued

	Placebo (n=160)			Sotatercept (n=163)			Sotatercept *versus* placebo difference in LS means at week 24 (95% CI); p-value
	Baseline Mean±SD (n)	Week 24 Mean±SD (n)	LS mean change from baseline at week 24 (95% CI) (n)	Baseline Mean±SD (n)	Week 24 Mean±SD (n)	LS mean change from baseline at week 24 (95% CI) (n)	
RC product (min)	61 203.00±22 153.80 (158)	63 280.80±21 710.40 (139)	1708.20 (−1476.00, 4891.80)	62 262.60±21 901.80 (154)	63 370.20±24 528.00 (150)	816.00 (−2279.40, 3910.80) (150)	−892.20 (−5260.20, 3475.80); p=0.6879
PA elastance ($mmHg \cdot mL^{-1}$)	1.35±0.53 (158)	1.39±0.63 (142)	0.05 (−0.02, 0.12) (142)	1.39±0.65 (155)	0.98±0.49 (150)	−0.37 (−0.44, −0.30) (150)	−0.42 (−0.52, −0.32); p<0.0001
RV work (g·m)	3.64±1.33 (160)	3.56±1.31 (144)	−0.14 (−0.30, 0.02) (144)	3.69±1.39 (163)	2.75±1.19 (154)	−0.99 (−1.14, −0.84) (154)	−0.85 (−1.07, −0.64); p<0.0001
RV power ($mmHg \cdot L \cdot min^{-1}$)	139.42±49.12 (160)	135.77±49.48 (144)	−5.81 (−11.79, 0.16) (144)	142.19±49.43 (163)	104.84±43.13 (154)	−38.51 (−44.24, −32.77) (154)	−32.70 (−40.86, −24.53); p<0.0001
S_{vO_2} (%)	67.44±7.93 (154)	66.42±9.35 (137)	−0.98 (−2.23, 0.27) (137)	66.80±7.10 (152)	69.94±7.38 (143)	2.86 (1.63, 4.09) (143)	3.84 (2.11, 5.57); p<0.0001

CE: cardiac efficiency; CI: cardiac index; CO: cardiac output; DBP: diastolic blood pressure; dPAP: diastolic pulmonary artery pressure; HR: heart rate; LS: least squares; mPAP: mean pulmonary artery pressure; mRAP: mean right atrial pressure; PA: pulmonary artery; PA compliance: pulmonary artery compliance; PAWP: pulmonary artery wedge pressure; PVR: pulmonary vascular resistance; RC: resistance–compliance; RV: right ventricular; SBP: systolic blood pressure; sPAP: systolic pulmonary artery pressure; SV: stroke volume; SVI: stroke volume index; SVR: systemic vascular resistance; S_{vO_2}: mixed venous oxygen saturation.

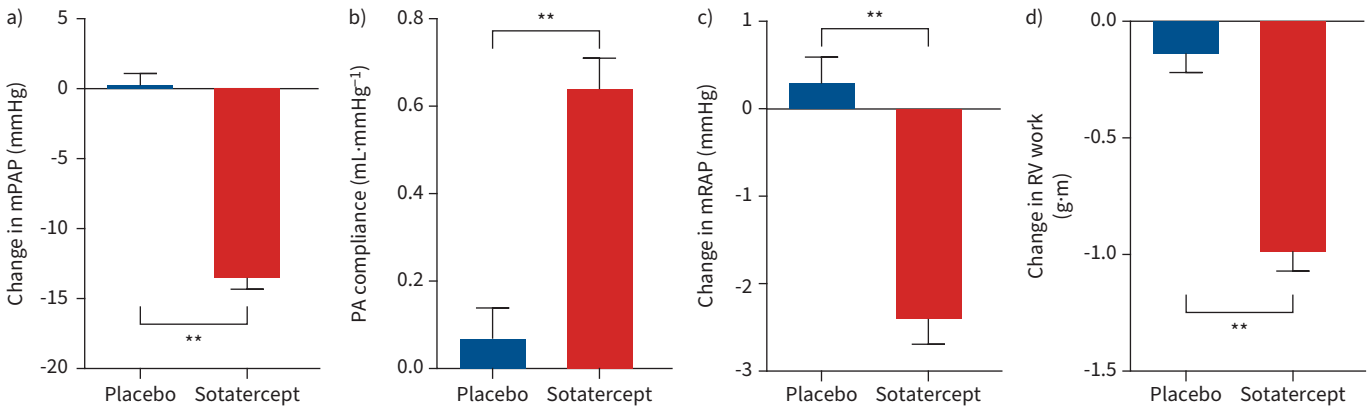

FIGURE 1 Least squares mean change from baseline in right heart catheterisation parameters of a) mean pulmonary artery pressure (mPAP), b) pulmonary artery (PA) compliance, c) mean right atrial pressure (mRAP), and d) right ventricle (RV) work at week 24 presented by treatment group. Bars represent standard errors. **: p<0.0001, sotatercept *versus* placebo.

Correlation analyses: change in 6MWD associations with RHC and ECHO parameters

The between-group change from baseline in 6MWD at week 24 was 39.89 m (95% CI 24.92 to 54.86 m; p<0.0001) in the overall analysis cohort based on the ANCOVA model.

Correlations between changes from baseline in RHC and ECHO variables with changes from baseline in 6MWD at week 24 are shown in supplementary tables S6 and S7, respectively. The most notable associations were moderate (CC ⩾0.400) negative correlations between change from baseline in 6MWD with change from baseline in mPAP (CC −0.418) and PVR (CC −0.408) (figure 5). Moderate (CC <0.400 but ⩾0.350) correlations also were seen between improvements in 6MWD and RHC-derived sPAP (CC −0.387), diastolic PA pressure (CC −0.398), PA elastance (CC −0.382) and cardiac efficiency (CC 0.358) (supplementary table S6). By comparison, the correlations between 6MWD and ECHO parameters were generally weak (all CC <0.300 for overall cohort).

Discussion

In this exploratory *post hoc* analysis of the phase 3 STELLAR trial, 24 weeks of treatment with the activin signalling inhibitor sotatercept improved haemodynamic and ECHO parameters in participants with PAH who were pre-treated with approved medications. Notably, the RHC findings showed a significant reduction from baseline in mPAP of 13.9 mmHg with sotatercept *versus* placebo, along with reduced RV work and RV power, improved cardiac efficiency, lowered mean RA pressure, and improved NT-proBNP levels. ECHO revealed no change in TAPSE, but a significant increase in TAPSE/sPAP ratio, reflecting improved RV–PA coupling, increased RVFAC, and a decline in RV volumes as well as a decline in the degree of tricuspid regurgitation. Collectively, these findings suggest that sotatercept treatment exerts beneficial effects on right heart function and dimensions by reducing PA pressure and RV workload in patients with PAH (see graphical abstract).

The substantial reduction in mPAP demonstrated that sotatercept effectively improved elevated pressures within the pulmonary circulation, a hallmark feature of PAH. Sotatercept treatment also led to a significant decrease in PVR of −255 dyn·s·cm^{-5} with sotatercept *versus* placebo, indicating reduced resistance to pulmonary blood flow. Although yet unproven in human disease, these findings may reflect partial reverse remodelling of the pulmonary arteries, the proposed mechanism of action for sotatercept [7]. Sotatercept treatment demonstrated significant improvements in PA compliance and PA elastance, both reflecting improved PA distensibility and reduced RV afterload. RV work and RV power decreased, indicating a reduction in the workload and energy expenditure of the RV with enhanced cardiac efficiency.

In the overall study population, sotatercept treatment was not associated with changes in cardiac output or cardiac index, respectively. The participants in this study were pre-treated with approved PAH medications, most of them with double or triple combination therapies, and entered the study with a mean cardiac index ⩾2.5 L·min^{-1}·m^{-2}. In subgroup analyses of participants who started with baseline cardiac index <2.5 L·min^{-1}·m^{-2}, treatment with sotatercept led to a numerical but non-significant increase in cardiac output, although not always achieving a value falling within the normal range at week 24. Despite an

TABLE 4 Baseline, week 24, and change from baseline values in echocardiography parameters across the placebo and sotatercept groups

	Placebo (n=133)			Sotatercept (n=142)			Sotatercept *versus* placebo difference in LS means at week 24 (95% CI); p-value
	Baseline Mean±SD (n)	Week 24 Mean±SD (n)	LS mean change from baseline at week 24 (95% CI) (n)	Baseline Mean±SD (n)	Week 24 Mean±SD (n)	LS mean change from baseline at week 24 (95% CI) (n)	
sPAP (mmHg)	76.32±21.75 (120)	76.50±21.46 (112)	−0.25 (−3.98, 3.49) (112)	73.66±24.82 (122)	58.99±24.94 (79)	−18.76 (−22.99, −14.53) (79)	−18.52 (−24.07, −12.96); p<0.0001
eRAP (mmHg)	6.80±2.48 (148)	7.33±2.74 (133)	0.42 (0.05, 0.79) (133)	6.80±2.46 (153)	5.61±1.57 (142)	−1.26 (−1.61, −0.91) (142)	−1.68 (−2.18, −1.18); p<0.0001
RVFAC (%)	29.28±8.94 (151)	30.24±9.26 (139)	1.42 (−0.04, 2.88) (139)	28.45±8.81 (155)	32.19±9.94 (148)	3.46 (2.04, 4.89) (148)	2.04 (0.03, 4.05); p=0.0462
RVA-ED (cm^2)	30.22±9.51 (151)	30.77±9.91 (139)	0.86 (−0.08, 1.80) (139)	30.69±8.87 (155)	25.93±8.75 (148)	−4.45 (−5.37, −3.54) (148)	−5.31 (−6.60, −4.02); p<0.0001
RVA-ES (cm^2)	21.75±8.70 (151)	21.85±8.62 (139)	0.26 (−0.53, 1.05) (139)	22.26±8.07 (155)	17.91±7.82 (148)	−4.13 (−4.90, −3.37) (148)	−4.39 (−5.47, −3.31); p<0.0001
RVEDA/LVEDA	0.44±0.17 (127)	0.47±0.23 (107)	0.02 (−0.01, 0.05) (107)	0.47±0.26 (118)	0.38±0.13 (115)	−0.06 (−0.09, −0.04) (115)	−0.08 (−0.12, −0.05); p<0.0001
RVESA/LVESA	1.06±0.59 (127)	1.19±0.92 (107)	0.04 (−0.06, 0.15) (107)	1.16±0.81 (118)	0.83±0.52 (115)	−0.27 (−0.37, −0.17) (115)	−0.32 (−0.46, −0.17); p<0.0001
TAPSE (mm)	19.19±4.17 (150)	18.84±4.28 (141)	−0.42 (−1.01, 0.17) (141)	19.07±4.44 (156)	18.69±4.05 (148)	−0.34 (−0.91, 0.23) (148)	0.08 (−0.73, 0.88); p=0.8483
TAPSE/sPAP ($mm \cdot mmHg^{-1}$)	0.28±0.12 (117)	0.27±0.12 (111)	−0.01 (−0.04, 0.01) (111)	0.29±0.14 (119)	0.38±0.19 (76)	0.11 (0.08, 0.14) (76)	0.12 (0.09, 0.16); p<0.0001
TRG (mmHg)	69.37±21.84 (127)	68.75±19.69 (119)	−0.46 (−3.84, 2.92) (119)	67.05±24.81 (126)	52.86±24.19 (85)	−17.00 (−20.87, −13.13) (85)	−16.54 (−21.58, −11.50); p<0.0001
Peak TRV ($m \cdot s^{-1}$)	4.11±0.66 (127)	4.10±0.61 (119)	−0.00 (−0.11, 0.10) (119)	4.03±0.75 (126)	3.55±0.79 (85)	−0.56 (−0.68, −0.44) (85)	−0.56 (−0.72, −0.40); p<0.0001
LVEF (%)	68.47±7.11 (129)	69.07±6.69 (109)	−0.22 (−1.48, 1.04) (109)	68.74±6.47 (119)	66.57±7.24 (117)	−2.28 (−3.51, −1.05) (117)	−2.06 (−3.79, −0.33); p=0.0200
LVEDVI ($mL \cdot m^{-2}$)	39.12±11.26 (129)	37.62±11.17 (109)	−0.65 (−2.43, 1.14) (109)	38.64±12.64 (119)	37.33±11.66 (117)	−1.54 (−3.28, 0.21) (117)	−0.89 (−3.35, 1.57); p=0.4774
LVESVI ($mL \cdot m^{-2}$)	12.48±5.20 (129)	11.71±4.49 (109)	−0.31 (−1.12, 0.50) (109)	12.37±5.60 (119)	12.66±5.55 (117)	0.33 (−0.46, 1.12) (117)	0.65 (−0.47, 1.76); p=0.2548

eRAP: estimated right atrial pressure; LS: least squares; LVEDVI: left ventricular end-diastolic volume index; LVEF: left ventricular ejection fraction; LVESVI: left ventricular end-systolic volume index; sPAP: systolic pulmonary artery pressure; RVA-ED: right ventricular area in end-diastole; RVA-ES: right ventricular area in end-systole; RVESA/LVESA: ratio of right ventricular end-systolic area to left ventricular end-systolic area; RVEDA/LVEDA: ratio of right ventricular end-diastolic area to left ventricular end-diastolic area; RVFAC: right ventricular fractional area change; TAPSE: tricuspid annular plane systolic excursion; TRG: tricuspid regurgitation gradient; peak TRV: peak tricuspid regurgitation velocity.

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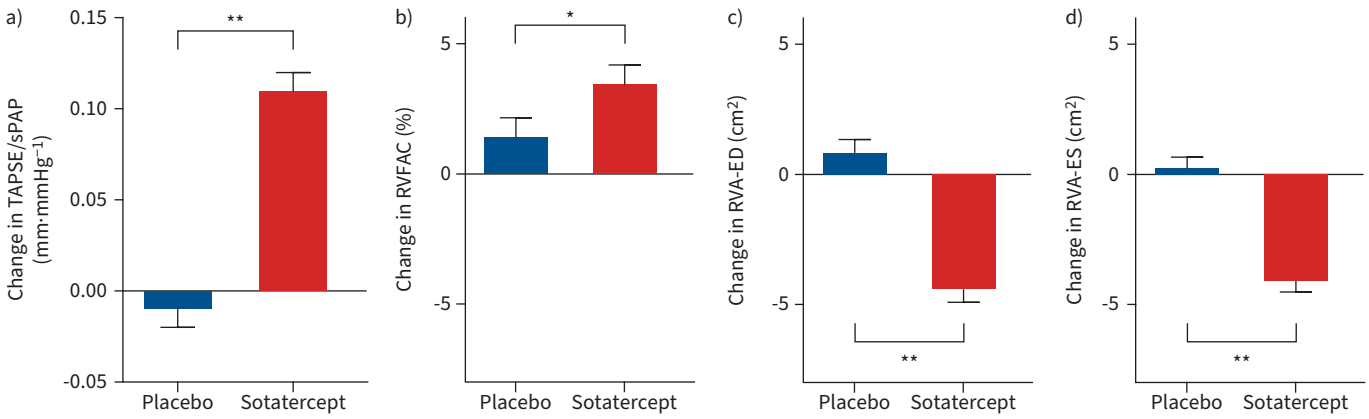

FIGURE 2 Least squares mean change from baseline in echocardiography parameters of a) tricuspid annular plane systolic excursion (TAPSE) to systolic pulmonary artery pressure (sPAP) ratio, b) right ventricular fractional area change (RVFAC), c) right ventricular area in end-diastole (RVA-ED), and d) right ventricular area in end-systole (RVA-ES) at week 24 presented by treatment group. Bars represent standard errors. **: p<0.0001, sotatercept *versus* placebo; *: p<0.050, sotatercept *versus* placebo.

unchanged cardiac output, a significant improvement in mixed venous oxygen saturation of 3.8% was seen with sotatercept *versus* placebo, at least in part due to increased haemoglobin levels.

The absence of a pronounced effect on cardiac output distinguishes sotatercept from other treatments currently approved for PAH, which decrease PVR at least partially by increasing pulmonary blood flow while having comparably modest effects on PA pressure [12–15]. The increase in cardiac output observed with currently approved PAH medication is probably related (at least in part) to their systemic vasodilatory effects, which leads to a reflex increase in cardiac output [17]. With sotatercept, systolic blood pressure and systemic vascular resistance increased, indicating that the drug did not cause systemic vasodilation but perhaps did cause some systemic vasoconstriction.

The ECHO findings expand on previous observations from the PULSAR study and laboratory biomarkers of RV strain also complement the changes seen during RHC [10]. TAPSE, a marker of longitudinal RV

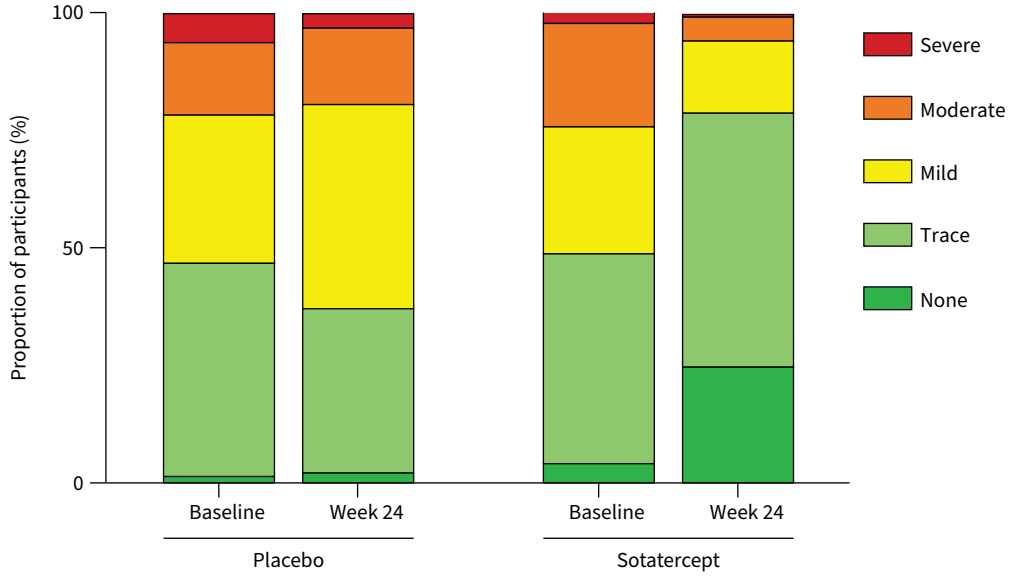

FIGURE 3 Proportion of participants (%) with various tricuspid regurgitation degree ratings at baseline and week 24 presented by treatment group.

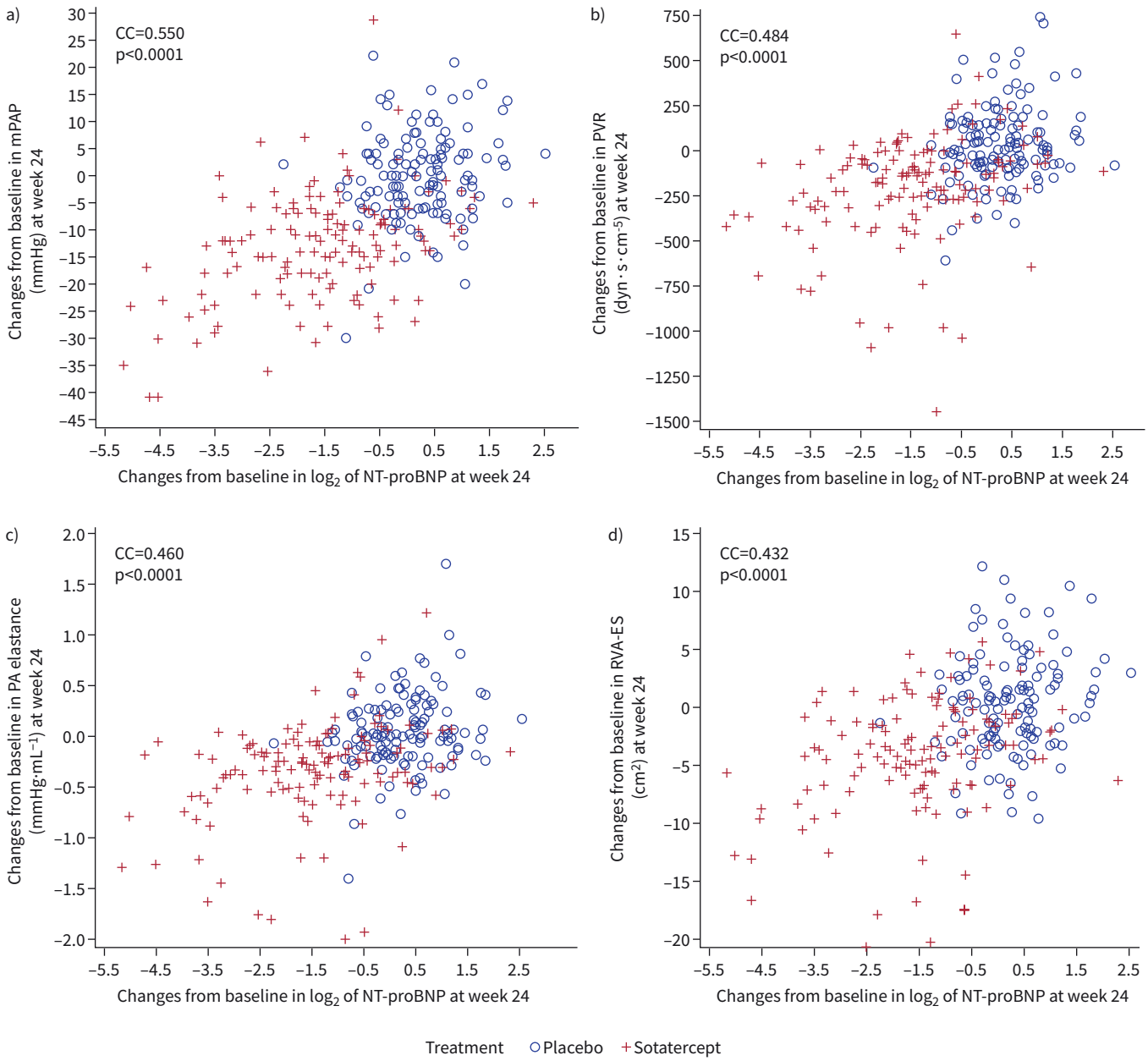

FIGURE 4 Scatterplots showing change from baseline in a) mean pulmonary artery pressure (mPAP), b) pulmonary vascular resistance (PVR), c) pulmonary artery (PA) elastance, and d) right ventricular area in end-systole (RVA-ES) and change from baseline in $\log_2$ of N-terminal pro-brain natriuretic peptide (NT-proBNP) at week 24. CC: correlation coefficient.

contractility, did not change with sotatercept treatment. However, due to the profound reduction in sPAP, the TAPSE/sPAP ratio improved substantially. The TAPSE/sPAP ratio has recently emerged as a prognostically important non-invasive measure of RV–PA coupling, RV afterload and load-dependent contractility [18, 19]. RVFAC improved and the RV size declined both in absolute numbers and as a comparison to the LV. Tricuspid regurgitation improved as well, reflecting less RV volume load, and probably contributing to improved PA–RV coupling [20].

LV end-systolic and LV end-diastolic volumes remained unchanged following treatment with sotatercept *versus* placebo but with a small decrease in LVEF (−2.06%; p=0.020). While the observed reduced LVEF could reflect normal variability or measurement error, there is the possibility that sotatercept treatment may lead to a slight decrease in LV systolic function, presumably related to the observed increase in systemic

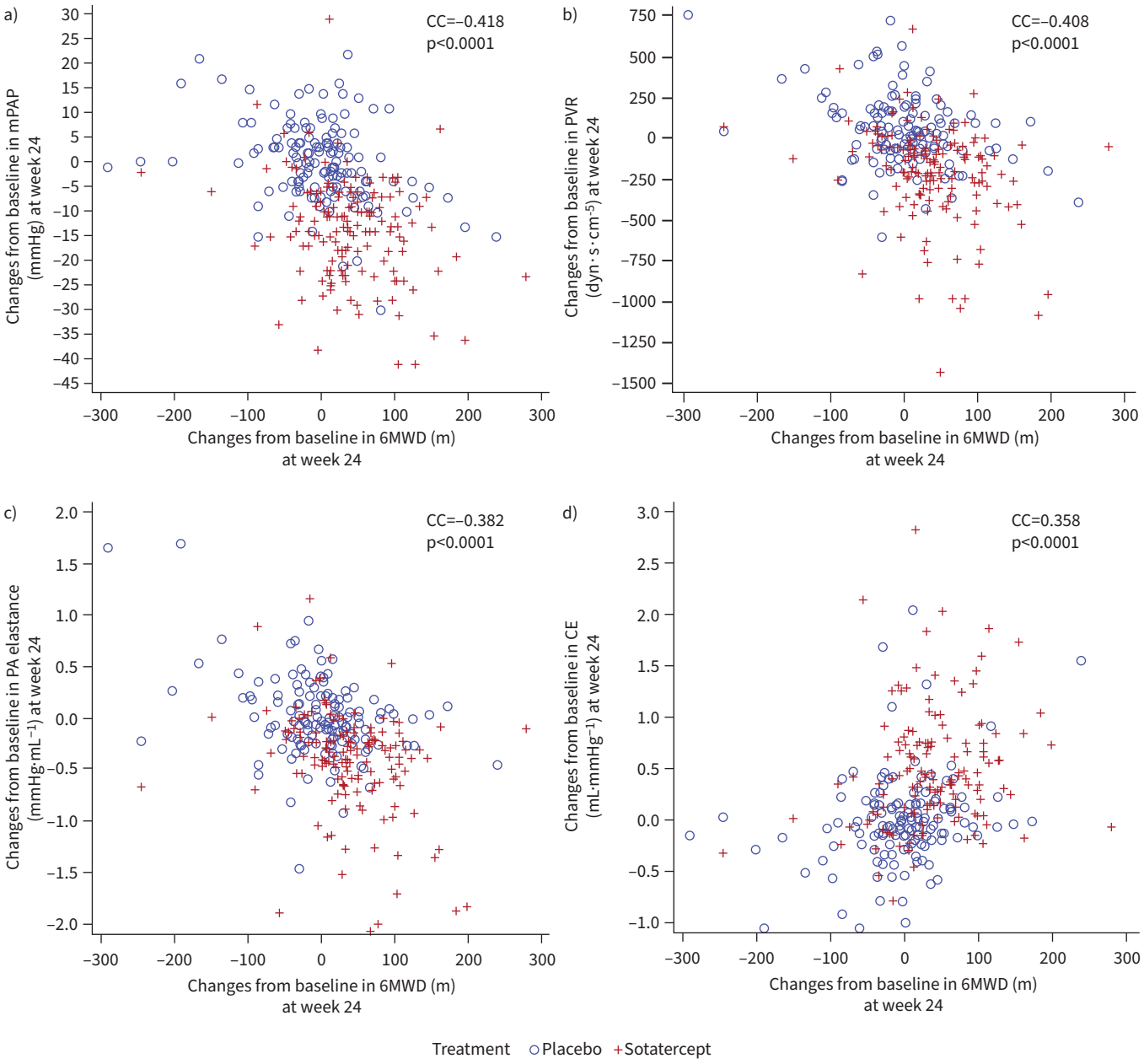

FIGURE 5 Scatterplots showing change from baseline in a) mean pulmonary artery pressure (mPAP), b) pulmonary vascular resistance (PVR), c) pulmonary artery (PA) elastance, d) cardiac efficiency (CE) and change from baseline in 6-min walk distance (6MWD) at week 24. CC: correlation coefficient.

vascular resistance. A reduction in LVEF was not observed in the phase 2 PULSAR study and its open-label extension, which followed participants for up to 24 months [10]. In the STELLAR trial, participants who received sotatercept had maintained heart rate, cardiac output and SV, a lower PA wedge pressure, higher mixed venous oxygen saturation, and improved NT-proBNP, all reassuring findings. Nevertheless, the effects of sotatercept on cardiac function need to be explored further, especially given preclinical data suggesting a role of activin signalling in myocardial regeneration [21]. The long-term safety of sotatercept including its effects on RV and LV function is currently being investigated in ongoing studies.

Sotatercept treatment resulted in a significant reduction in NT-proBNP, a global marker of RV strain, which has emerged as a robust predictor of outcomes in patients with PAH [22–24]. Changes in NT-proBNP were correlated with changes in various RHC and ECHO parameters. Most of the variables that were moderately or strongly correlated with changes in $\log_2$ NT-proBNP are either directly or

indirectly linked to PA pressure and right heart dimensions, which underscores the therapeutic role of reducing elevated PA pressures in patients with PAH. Changes in 6MWD, the primary outcome measure of the STELLAR study, were correlated with changes in mPAP, PVR, PA elastance and cardiac efficiency. We assume that the improvements in exercise capacity observed with sotatercept reflect the sum effects of decreased RV afterload, improved cardiac performance and increased oxygen delivery, with the latter mainly caused by increased haemoglobin values.

There are several limitations that should be considered when interpreting the results of this analysis. First, this was a *post hoc* exploratory analysis which aimed to generate hypotheses and explore potential associations rather than establish definitive causal relationships. Multiple tests were performed without correcting for multiplicity which can inflate the type 1 error rate, increasing the likelihood of detecting a significant effect due to chance. Therefore, the findings should be interpreted with caution, and further confirmatory studies are needed to validate the observed effects. This analysis also may have been limited by a relatively small sample size, which can affect the statistical power and generalisability of the results. It is important to acknowledge that this analysis focused on a narrow population of participants with group 1 PAH and WHO functional class II–III symptomatic disease, thus the results may not be directly applicable to other PAH populations or disease stages. Although efforts were made to account for confounding factors in this analysis using an ANCOVA model, there may still be unmeasured or residual confounding variables that may have influenced the observed results. While RHC and ECHO are valuable tools for assessing cardiac function, they both are limited by inter-observer variability and technical challenges in obtaining accurate measurements. Finally, the associations reported here are based on data accumulated over 24 weeks of observation, thus providing only a short-term assessment of the associations in these parameters over time.

In conclusion, this *post hoc* exploratory analysis of the phase 3 STELLAR trial provides insights into the effects of sotatercept on haemodynamics and right heart function in patients with PAH. The findings suggest that sotatercept profoundly reduces PA pressures, thereby improving PA compliance and RV–PA coupling, reducing tricuspid regurgitation and cardiac workload, and improving right heart dimensions and function. Together with the clinical benefits observed with sotatercept in the STELLAR trial, these results underscore the therapeutic relevance of reducing PA pressures in the management of patients with PAH.

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This study was prospectively registered at ClinicalTrials.gov as NCT04576988. The data sharing policy, including restrictions, of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA, is available at http://engagezone.msd.com/ds_documentation.php. Requests for access to the study data can be submitted through the Engage Zone site or *via* email to dataaccess@merck.com.

Conflict of interest: R. Souza has served as consultant for Acceleron Pharma, Inc., Bayer Healthcare Pharmaceuticals Inc. and Janssen Biotech, Inc. (fees paid to self). D.B. Badesch has received grants/contracts from Acceleron Pharma, Inc., Merck & Co., Inc. (Rahway, NJ, USA), Altavant and United Therapeutics (all paid to institution, clinical trial), received payment as consultant from Acceleron Pharma, Inc., Merck & Co., Inc. (Rahway, NJ, USA) and Aerovate Inc. (paid through institution), and the author's spouse or partner has stock in Johnson & Johnson Health Care Systems Inc. H.A. Ghofrani has served as consultant for Aerovate, Altavant, Bayer Healthcare, Gossamer Bio, Janssen Diagnostics, LLC, Merck & Co., Inc. (Rahway, NJ, USA) and Pfizer (fees paid to self), and is an employee of Justus Liebig University Giessen, Germany. J.S.R. Gibbs has served a consultant for Acceleron Pharma and Merck & Co., Inc. (Rahway, NJ, USA) (fees paid to self), served in data and safety monitoring boards for Actelion Pharmaceuticals, Fundação Bial, GossamerBio and Merck & Co., Inc. (Rahway, NJ, USA) (fees paid to self), served in end point review committee for Actelion Pharmaceuticals, Aerovate, Janssen Biotech, Pfizer Pharma GMBH and United Therapeutics Corporation (fees paid to self), and served as Chair of ERN-Lung Functional Committee Patient Reported Outcomes for ERN Lung. M. Gomberg-Maitland has served as a consultant for Acceleron Pharma and Merck & Co., Inc. (Rahway, NJ, USA), Aerami, Bayer HealthCare Pharmaceuticals Inc., Janssen Biotech, Keros and United Therapeutics Corporation (fees paid to self), has received grant/contract from Aerovate, Altavant, Acceleron Pharma and Merck & Co., Inc. (fees paid to institution), and the author's spouse is

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